國 立 清 華 大 學

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碩 士 論 文

Master's Thesis

基於自我確定性評估與熵門控的知識檢索

SEEK: Self-certainty Evaluated Entropy-gated Knowledge-retrieval

系所別： 資訊系統與應用研究所

Dept. / Grad. Inst.: Institute of Information Systems and Applications

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中 華 民 國 一一五 年 七 月

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Abstract

This study proposes SEEK (Self-certainty Evaluated Entropy-gated Knowledge-retrieval), a lightweight adaptive Retrieval-Augmented Generation (RAG) framework for open-domain question answering. Standard RAG systems apply a fixed retrieval strategy to every query, which can inject noise on simple questions and miss intermediate evidence on multi-hop questions. Existing adaptive methods either route at a coarse question level, probe uncertainty at a noisy token level, or learn expensive sub-question policies through tree search and preference optimisation.

SEEK uses a compact KG-triple decomposer trained with a single LoRA SFT pass on a 12K mixture of MuSiQue, 2WikiMultihopQA, and PopQA examples. The final system no longer treats entropy as a single hard skip gate. Instead, it generates both a direct answer and a decomposed answer, preserves original-question evidence inside decomposed sub-question contexts, and applies a training-free verifier with a conservative type-aware fallback. This late-routing design preserves the interpretability of entropy-based methods while avoiding the brittleness of a global threshold.

Under the honest wiki-21M retrieval protocol aligned with DeepRAG, SEEK reaches an average of 40.52 EM and 47.61 F1, exceeding DeepRAG's 38.45 EM and 45.75 F1. SEEK improves over DeepRAG on 2WikiMultihopQA (50.00 EM), MuSiQue (24.70 EM), and PopQA (51.90 EM), while remaining behind on HotpotQA (35.50 EM versus 40.70 EM). Analysis shows that direct and decomposed paths are complementary, that oracle direct-vs-decompose routing still offers 3.8 to 6.1 EM of headroom, and that entropy is useful as a sub-question retrieval signal but not as a correctness guarantee.

Keywords: Retrieval-Augmented Generation, Adaptive Retrieval, Multi-hop Question Answering, Entropy, Verifier Routing, Large Language Models

摘要

本研究提出 SEEK (Self-certainty Evaluated Entropy-gated Knowledge-retrieval)，一個用於開放領域問答的輕量化自適應檢索增強生成 (Retrieval-Augmented Generation, RAG) 框架。傳統 RAG 對所有問題採用固定檢索策略：簡單問題可能因不必要的檢索引入雜訊，多跳問題則可能因單次檢索無法找到中介證據而回答失敗。既有自適應方法不是在問題層級做過於粗糙的路由，就是在詞元層級使用嘈雜的不確定性訊號，或像 DeepRAG 一樣透過樹搜尋與偏好最佳化學習昂貴的子問題決策策略。

SEEK 使用一個以 KG-triple 格式表示的分解器，透過單次 LoRA SFT 在 12K 筆 MuSiQue、2WikiMultihopQA 與 PopQA 混合資料上訓練。最終版本不再把熵值當成單一硬性 skip gate，而是同時產生 direct answer 與 decomposed answer，並將 original-question evidence 加回 decomposed sub-question context，最後由 training-free verifier 搭配保守的 type-aware fallback 選擇最終答案。這種 late-routing 設計保留了熵訊號的可解釋性，同時避免單一全域閾值在跨資料集時的不穩定。

在與 DeepRAG 對齊的 honest wiki-21M 檢索設定下，SEEK 平均達到 40.52 EM 與 47.61 F1，高於 DeepRAG 的 38.45 EM 與 45.75 F1。分資料集來看，SEEK 在 2WikiMultihopQA 達到 50.00 EM、MuSiQue 達到 24.70 EM、PopQA 達到 51.90 EM，皆高於 DeepRAG；HotpotQA 則爲 35.50 EM，仍低於 DeepRAG 的 40.70 EM。分析顯示 direct path 與 decomposition path 具有互補性，oracle direct-vs-decompose routing 仍有 3.8 到 6.1 EM 的提升空間，而 entropy 適合作爲 sub-question retrieval signal，但不能被視爲 correctness guarantee。

關鍵字：檢索增強生成、自適應檢索、多跳問答、熵、驗證器路由、大型語言模型

Acknowledgements

I would like to express my sincere gratitude to my advisor for the guidance and support throughout this research. I also thank the members of the Intelligent Knowledge Management Lab at National Tsing Hua University, the thesis oral examination committee, and all those who provided computational resources and helpful discussions. Finally, I thank my family and friends for their continuous encouragement.

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Chapter 1 Introduction

1.1 Background and Motivation

Large language models (LLMs) often hallucinate when queried about recent, obscure, or proprietary information outside their training scope. Retrieval-Augmented Generation (RAG) addresses this by fetching relevant external evidence, reducing factually unsupported answers [1], [2]. However, applying RAG indiscriminately creates two problems. For simple questions, unnecessary retrieval can inject distracting evidence and increase latency. For complex multi-hop questions, a single retrieval step often fails to surface the intermediate entities needed for reasoning.

Adaptive RAG therefore needs to decide not only *whether* retrieval is useful, but also *when to trust decomposition*. Decomposition can sharpen retrieval queries, but it can also remove the original question's comparison targets, entity pairs, or disambiguating clues. This thesis therefore frames adaptive RAG as a *decomposition-routing problem*: the system should control retrieval at the sub-question level while preserving the original-question path as a competing candidate.

1.2 Research Objectives

This thesis introduces SEEK (Self-certainty Evaluated Entropy-gated Knowledge-retrieval), a lightweight adaptive RAG framework built around training-free verifier routing. The objectives are:

- Build a compact KG-triple decomposer that converts multi-hop questions into atomic `entity >> relation` sub-questions.
- Use token-level entropy as an interpretable confidence signal, while showing that entropy alone is insufficient as a global hard router.

- Generate both direct and decomposed candidate answers, preserve original-question evidence inside decomposed steps, and select the final answer with a training-free verifier based on confidence, answer type, decomposition stability, and candidate agreement.
- Evaluate the final system under the honest wiki-21M retrieval protocol used by DeepRAG, showing where a lightweight verifier can match or exceed a learned retrieval policy and where routing headroom remains.

Chapter 2 Related Work

2.1 Adaptive RAG

Retrieval-Augmented Generation (RAG) [1] augments a parametric language model with a non-parametric retriever, allowing the model to condition its generation on external evidence rather than relying only on parameters learned during pretraining. A typical RAG pipeline can be described as five stages:

1. **Indexing.** The source corpus is segmented into passages, and each passage is stored in a form that supports retrieval, such as sparse lexical indexes or dense vector representations.

2. **Query manipulation.** The input question may be rewritten, expanded, normalised, or decomposed so that the retriever can better match it with relevant evidence.

3. **Retrieval.** The system selects candidate passages from the indexed corpus, commonly using sparse lexical matching, dense vector similarity, or a hybrid retrieval strategy.

4. **Post-retrieval processing.** The initially retrieved passages may be filtered, reranked, compressed, or otherwise refined before they are passed to the generator.

5. **Generation.** The language model conditions on the selected evidence and produces the final answer, ideally grounding its output in the retrieved passages.

This retrieve-then-generate paradigm is now widely used to make language model outputs more factual in question answering, explanation writing, and other knowledge-intensive tasks. Its central advantage is that factual evidence can be updated externally without retraining the language model. For multi-hop question answering, however, the quality of the retrieved evidence depends strongly on the query that is issued at each step: a single original question may not contain the intermediate entity needed to retrieve the

next supporting passage.

Despite its benefits, the default "always retrieve" policy has two well-documented failure modes:

- **Noise injection on easy queries.** For questions the model already knows (e.g. "Who wrote *Hamlet*?"), retrieved passages introduce distractor entities that can override correct parametric knowledge [2].
- **Insufficient evidence on hard queries.** For multi-hop questions, a single retrieval pass rarely surfaces all bridging entities, leading to incomplete or compositionally wrong answers [3].

These two failure modes motivate adaptive retrieval: deciding *when* retrieval is useful, *how much* evidence to fetch, and what signal should determine the gate.

Adaptive RAG methods decide per query (or per reasoning step) whether and how much to retrieve. They can be characterised along three axes: the **decision granularity** (question, sentence, token, or sub-question), the **signal** used to gate retrieval, and the **training cost** (training-free, SFT, RL, or DPO). Organising by the gating signal — the most consequential axis for SEEK — yields three families: *classifier-based*, *confidence-based*, and *reasoning- and policy-based* gating.

Classifier-based gating. The first family trains a dedicated discriminator that inspects the query and emits a retrieval decision.

Adaptive-RAG [4] routes each incoming query to one of three strategies based on its complexity: (i) *no retrieval* — the LLM answers from parametric memory only; (ii) *single-step retrieval* — one BM25 or dense retrieval pass followed by reading; and (iii) *multi-step retrieval* — iterative retrieve-and-reason (e.g. IRCoT [3]). The routing decision is produced by a small T5-Large classifier trained on *automatically derived labels*: each training question is labelled with the cheapest strategy whose prediction matches the gold answer, with ties broken by the dataset prior (single-hop datasets bias toward "no retrieval"; multi-hop datasets toward "multi-step"). Evaluated on Natural Questions, Triv-

iaQA, SQuAD, MuSiQue, HotpotQA, and 2WikiMultihopQA, Adaptive-RAG matches a uniform multi-step baseline while approximately halving the average retrieval count. Its chief limitation is that the gate fires *once per query*, before any reasoning has occurred: the classifier cannot react to mid-reasoning uncertainty, and the three labels are coarse — a five-hop and a two-hop question both receive "multi-step".

SKR [5] learns a binary "does the model know it?" classifier from the model's own correctness on a calibration set. At inference, a query is classified as "known" (skip retrieval) or "unknown" (retrieve), making SKR effectively a binary version of Adaptive-RAG: cheaper than reflection-token methods, but unable to distinguish single-step from multi-step needs. Both classifier-based methods require an additional trained head and, more fundamentally, commit to a decision before the model has begun to reason.

Confidence- and uncertainty-based gating. The second family replaces the external classifier with an *intrinsic* uncertainty signal read off the generation process, and is typically training-free.

FLARE [6] pushes the gating granularity from question level to sentence level. At each generation step it first drafts a tentative next sentence; if any token in that sentence carries a probability below a threshold θ, the low-probability span is masked, the sentence is re-issued as a retrieval query, and generation is restarted with the retrieved evidence. The method is training-free and language-agnostic, but token probability is a noisy proxy for factual uncertainty (common-word low-probability spans trigger spurious retrievals), and sentence-level rewriting interacts poorly with chain-of-thought, whose natural reasoning unit is a sub-question rather than a sentence.

DRAGIN [7] extends FLARE with *Real-time Information Need Detection* (RIND) and *Query Formulation based on Self-attention* (QFS). RIND scores each generated token by combining token entropy, attention weight to subsequent tokens, and semantic content, triggering retrieval when the score exceeds a threshold; QFS then builds the query from the most-attended tokens in the preceding context rather than the raw lookahead. DRAGIN is training-free and improves over FLARE on 2WikiMultihopQA, HotpotQA, and IIRC, but inherits FLARE's token-level granularity mismatch and adds attention-

extraction overhead.

SeaKR [8] taps the residual stream directly, estimating uncertainty by aggregating the variance of hidden states across decoding steps; high variance triggers retrieval. The authors argue that internal variance captures factual uncertainty more faithfully than output-layer entropy, and add a re-ranking step that prefers passages which reduce hidden-state variance. SeaKR is reported as state-of-the-art among training-free methods on Natural Questions, HotpotQA, and 2WikiMultihopQA. Across this family, however, the gate ultimately depends on a hand-set threshold over a noisy uncertainty proxy — the central tension SEEK addresses by using entropy as one verifier signal rather than as a single hard global router.

Reasoning- and policy-based gating. The third family folds the retrieval decision into the model's own generation, learning it as part of a reasoning policy.

Self-RAG [9] trains the LLM to emit special *reflection tokens* — `[Retrieve]` (gate), `[IsRel]` (passage relevance), `[IsSup]` (span support), and `[IsUse]` (utility) — by fine-tuning on GPT-4-distilled traces, then performs token-conditioned tree decoding at inference. It outperforms retrieve-then-read baselines on TriviaQA, PopQA, and ALCE long-form generation, but requires a costly distillation corpus and a custom decoder that interleaves reflection-token sampling with content generation.

DeepRAG [10] models retrieval-augmented reasoning as a Markov Decision Process: at each sub-query the agent chooses between answering from parametric memory and issuing a retrieval call. Training has two stages — *Stage I* uses binary tree search to roll out both branches per sub-query, producing a balanced SFT corpus that teaches both "skip" and "retrieve" patterns; *Stage II* (*Chain of Calibration*) applies DPO-style preference learning over the traces. Trained on 4K HotpotQA + 1K 2WikiMultihopQA examples ($\approx$10K via tree augmentation) and evaluated out-of-distribution on PopQA, CAG, WebQuestions, and MuSiQue, DeepRAG achieves the strongest sub-question-level gating to date — but at substantial cost: tree search is $O(2^H)$ in the number of hops, the DPO stage requires careful pair construction, and the gate is a learned implicit policy with no externally interpretable score.

Orthogonal to the gating signal, a complementary line of work treats the retrieval *count* as a user-controllable knob: **Fast-or-Better** [11] exposes a Pareto frontier between latency and accuracy rather than a single operating point, underscoring that adaptive retrieval is fundamentally a cost–accuracy trade-off. SEEK positions itself within this family in spirit — sub-question-level reasoning like DeepRAG — but replaces the learned policy with a training-free verifier that combines entropy, decomposition stability, and candidate agreement.

Table 2.1 summarises the key differences between SEEK and existing adaptive RAG methods along the three axes introduced above. SEEK occupies the under-explored region of *late verifier-based routing with interpretable, training-free signals*: only a single SFT pass is needed for the decomposer, and the final route is decided by transparent query-level diagnostics rather than a learned policy.

Table 2.1: Comparison of adaptive RAG methods along granularity, gating signal, interpretability, training cost, and decision unit.

Method	Decomposer	Gating Signal	Interpretable	Training Cost	G
Adaptive-RAG [4]	None	Query classifier	Partial	Low (T5-Large)	
FLARE [6]	None	Token probability	Yes	None	
Self-RAG [9]	None	Reflection tokens	Partial	High (distill)	
DRAGIN [7]	None	Entropy + attention	Yes	None	
SKR [5]	None	Binary self-knowledge	Partial	Low	
SeaKR [8]	None	Hidden-state variance	No	None	
DeepRAG [10]	SFT	Learned policy (SFT+DPO)	No	High	
SEEK (Ours)	SFT	Entropy-aware verifier	**Yes**	**Low** (SFT only)	**Sub**

2.2 Reasoning and Knowledge Boundaries in RAG

A parallel thread focuses on *how* to retrieve and reason once retrieval is deemed necessary, independent of the gate.

The foundational technique is **IRCoT** [3], which interleaves chain-of-thought reasoning with retrieval: after each generated CoT sentence, the partial reasoning becomes the next query. IRCoT is the canonical "multi-step" implementation referenced by Adaptive-RAG and DeepRAG, but its retrieval count grows linearly in the CoT length, regardless of whether intermediate steps are factually difficult. **Question decomposition** [12] instead explicitly splits a multi-hop question into atomic sub-questions before answering, treating decomposition as a separate seq-to-seq problem. SEEK inherits this decomposition paradigm, but its final design does not make a hard pre-decomposition choice: it generates both direct and decomposed candidates, preserves original-question context for decomposed steps, and verifies which path is more reliable.

More recent work synthesises reasoning data to internalise the retrieve-and-reason loop. **Auto-RAG** [13] autonomously synthesises multi-turn retrieval–reasoning trajectories and fine-tunes on them, so that the model plans its own retrieval steps without hand-crafted prompts. **Search-o1** [14] embeds retrieval inside a large reasoning model in an agentic loop, augmenting o1-style long-chain reasoning with on-demand search and a document-reasoning module that distils retrieved passages before they re-enter the chain. **AirRAG** [15] activates the model's intrinsic reasoning by combining Monte Carlo Tree Search with self-consistency to explore alternative solution paths. These methods substantially strengthen the reasoning side of RAG, but — like DeepRAG — they rely on heavy data synthesis or search-time expansion; SEEK instead keeps the reasoning component a single lightweight SFT decomposer and concentrates its novelty in training-free verification.

Every adaptive method above ultimately rests on a model's ability to recognise its own *knowledge boundary* — the line between what it reliably knows and what it must look up. A dedicated line of work studies this capacity directly, and we group it into three strands: *self-knowledge elicitation*, *uncertainty estimation*, and the *interaction between the knowledge boundary and retrieval*.

Self-knowledge and its calibration. A first strand asks whether a model can introspect on what it knows. Kadavath et al. [16] show that LLMs are, to a useful degree, cali-

brated: a model can be prompted to predict the probability `P(True)` that its own answer is correct, and these self-judgements correlate with actual accuracy — though calibration degrades on out-of-distribution and adversarial questions. Yin et al. [17] construct a "SelfAware" benchmark of answerable and unanswerable questions and find that LLMs only partially "know what they don't know": self-knowledge can be elicited but lags well behind human performance. From the supervised side, Kapoor et al. [18] argue that models "must be taught to know what they don't know", demonstrating that lightweight calibration-tuning or probing substantially sharpens a model's estimate of its own uncertainty. Yin et al. [19] formalise the notion of a knowledge boundary itself, proposing to benchmark a model by probing the frontier of its reliable knowledge rather than by aggregate accuracy alone.

Uncertainty estimation as a boundary signal. A second strand asks how to *measure* the boundary at inference time. A naive token-probability or token-entropy signal conflates two kinds of uncertainty: *epistemic* (the model genuinely does not know) and *lexical* (many surface forms express the same fact). Kuhn et al. [20] introduce *semantic entropy*, which clusters multiple sampled answers by meaning before computing entropy, so that paraphrases of one correct answer do not register as uncertainty; Farquhar et al. [21] scale this idea and show in *Nature* that semantic entropy reliably detects confabulations across QA and biography generation. This line of work is the closest in spirit to SEEK's verifier signal: both read out the knowledge boundary from generation-time uncertainty. SEEK deliberately adopts the cheaper *token-level* entropy and combines it with decomposition stability and candidate agreement, avoiding the brittleness of an entropy-only hard gate.

The boundary under retrieval. A third strand studies the boundary specifically in the RAG setting. Ren et al. [22] show that retrieval augmentation not only injects evidence but also *reshapes* the boundary: relevant passages let a model answer questions it could not before, while irrelevant or distracting passages can push a previously-correct answer over the boundary into error — precisely the "noise injection on easy queries" failure mode of §2.1. This makes a per-step boundary estimate, rather than a one-shot query-level one, essential when retrieval and reasoning are interleaved.

These three strands form the conceptual foundation of SEEK. The failure modes of §2.1 — spurious retrieval on easy queries, missing retrieval on hard ones, and confident hallucination under a hard entropy threshold — are all symptoms of a poorly localised knowledge boundary. Where confidence-based methods read a single uncertainty proxy and policy-based methods bury the boundary inside a learned policy, SEEK makes the boundary inspectable through a late verifier: entropy, decomposition stability, and answer agreement jointly decide whether the direct or decomposed path should be trusted.

2.3 Evaluation Datasets

The evaluation of adaptive RAG hinges on datasets that span both single-hop and multi-hop complexity. This thesis uses HotpotQA, 2WikiMultihopQA, MuSiQue, and PopQA to cover both multi-hop reasoning and single-hop factual recall.

HotpotQA. HotpotQA [23] is the canonical multi-hop QA benchmark, containing 113K Wikipedia-based questions requiring 2-hop reasoning. Each example provides (a) the question, (b) the gold answer, (c) two gold-supporting paragraphs out of 10 distractors (in the `distractor` setting), and (d) sentence-level supporting facts. Questions fall into two compositional patterns: *bridge* (the answer to sub-question 1 is the subject of sub-question 2) and *comparison* (two entities are compared along a shared attribute). HotpotQA is a primary training source for DeepRAG and a standard evaluation set for nearly all adaptive RAG papers.

2WikiMultihopQA. 2WikiMultihopQA [24] constructs 192K multi-hop questions by leveraging Wikidata knowledge-graph paths, ensuring each question requires traversal of an explicit reasoning chain. It includes four question types (comparison, inference, compositional, and bridge-comparison) and provides KG-triple annotations alongside textual evidence. Compared to HotpotQA, 2WikiMultihopQA exhibits less surface-form leakage between the question and the supporting passages, making it a more reliable test of compositional reasoning.

MuSiQue. MuSiQue (**Mu**ltihop questions via **Si**ngle-hop **Que**stions) [25] addresses two pathologies of earlier datasets: (i) shortcut answers via single-hop reasoning, and (ii) weak distractor selection. MuSiQue is built by composing 2–4 single-hop questions into chains using strict connectivity criteria, yielding 25K questions of 2-hop, 3-hop, and 4-hop varieties. Importantly, MuSiQue provides the *decomposed* sub-questions as supervision, which makes it a natural training source for decomposer models such as the one used in SEEK. Its high distractor difficulty makes it the hardest of the four datasets in this thesis.

PopQA. PopQA [26] is a *single-hop* entity-centric QA dataset of 14K questions, each derived from a Wikidata subject–property–object triple. Critically, PopQA tags each entity with its *popularity* (measured by monthly Wikipedia page views), enabling stratified evaluation of factual recall on long-tail entities. Despite being single-hop, PopQA exposes the over-decomposition failure mode of decomposer-based pipelines: trained on multi-hop data, a decomposer may emit two-step plans even when one step suffices, harming both EM and efficiency.

Single-hop open-domain QA. For completeness, prior adaptive-RAG papers also evaluate on **Natural Questions (NQ)** [27], **TriviaQA** [28], and **SQuAD** [29]. These are single-hop datasets whose answers typically exist in a single Wikipedia paragraph, and they primarily test whether an adaptive method can correctly *skip* retrieval rather than orchestrate it. This thesis follows the DeepRAG protocol of evaluating on PopQA as the representative single-hop test set.

Chapter 3 Methodology

This chapter presents **SEEK** (**S**elf-certainty **E**valuated **E**ntropy-gated **K**nowledge-retrieval), a lightweight adaptive RAG framework that treats question decomposition as a routing problem. Earlier versions of SEEK used entropy as a hard skip-decomposition gate. The final system instead runs both a direct path and a decomposition path, preserves original-question evidence inside decomposed steps, and chooses between the candidate answers using a training-free verifier. Section 3.1 gives the end-to-end overview. Sections 3.3–3.6 describe the decomposition path, Section 3.7 describes the verifier, and Sections 3.8–3.9 give the training and implementation details.

3.1 Overview

Given a question q and an open-domain retrieval corpus $\mathcal{P}$, SEEK produces a final answer $\hat{a}$ through two candidate-generation paths followed by a verifier:

1. **Direct path.** The answerer directly answers the original question q with BM25 evidence retrieved from $\mathcal{P}$, producing a direct candidate a^{dir} and a token-level entropy score σ^{dir} over the generated answer span.

2. **Decomposition path.** The decomposer $\mathcal{D}_\theta$ maps q to ordered sub-questions $\{s_1, \ldots, s_n\}$. Each sub-question is answered sequentially, with placeholder substitution, entropy probing, conditional retrieval, original-question evidence union, and a final synthesis step that produces a^{dec}.

3. **Training-free verifier.** A deterministic verifier compares a^{dir} and a^{dec} using query-level signals: direct-answer confidence, decomposition stability, post-synthesis confidence, and answer agreement. It returns either a^{dir} or a^{dec} as $\hat{a}$.

The final operating point is the **training-free verifier with type-aware fallback**: it uses only signals naturally emitted during inference and does not require an additional classifier, DPO, tree search, or gold-answer supervision. Evidence-support variants are

evaluated as ablations in Chapter 4, but are not used in the main method because lexical support is not globally stable for compositional reasoning.

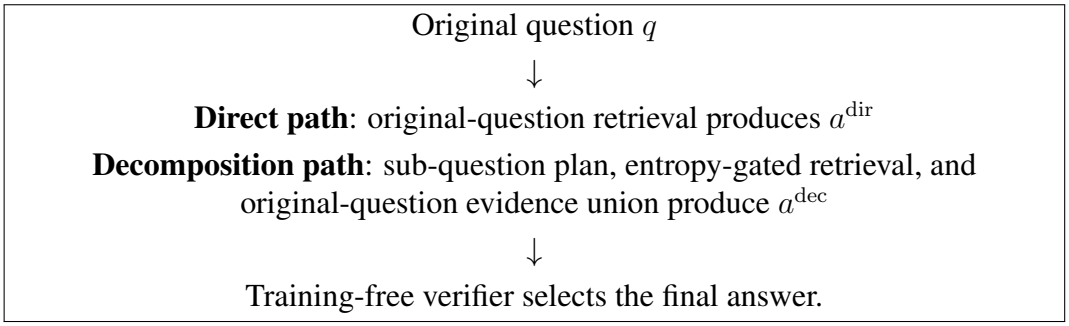


Fig. 3.1: SEEK pipeline: the system generates a direct answer and a decomposed answer, preserves original-question evidence inside the decomposition path, and uses a training-free verifier to select the final answer.

3.2 Problem Formulation

Let $\mathcal{M}_\theta$ be a pretrained language model, $r(q, \mathcal{P}, k)$ a BM25 retriever over an open-domain corpus $\mathcal{P}$, and $\mathcal{D}_\theta(q)$ a decomposer that emits sub-questions. A conventional retrieve-then-read system produces one answer from one path. SEEK instead constructs two candidates:

$$a^{\mathrm{dir}} = \mathcal{M}_\theta(q, r(q, \mathcal{P}, k)),$$

$$a^{\mathrm{dec}} = \mathrm{Synth}\left(q, \{(s_i, a_i)\}_{i=1}^{n}\right), \quad \{s_i\}_{i=1}^{n} = \mathcal{D}_\theta(q).$$

The final decision is a verifier v:

$$\hat{a} = v\left(q, a^{\mathrm{dir}}, a^{\mathrm{dec}}, z^{\mathrm{dir}}, z^{\mathrm{dec}}\right),$$

where z^{dir} and z^{dec} are inference-time diagnostic signals such as entropy, unresolved placeholders, and candidate agreement. Unlike DeepRAG [10], SEEK does not learn v through tree-search rollouts or preference optimisation. The verifier is a transparent rule

set designed to approximate the oracle choice between direct and decomposed reasoning.

3.3 Stage 1: Sub-Question Decomposer

3.3.1 Output Format

The decomposer emits one sub-question per line, prefixed with `#`$\langle i \rangle$`:`. Each sub-question is expressed as a compact knowledge-graph fragment:

$$\texttt{\#}i\texttt{:} \quad \langle \textit{entity} \rangle \;\; \texttt{>>} \;\; \langle \textit{relation} \rangle$$

When a sub-question depends on an earlier answer, the placeholder `#`j refers to step j. For example:

```
#1: X >> director
#2: #1 >> place of birth
```

At inference, placeholders are resolved sequentially: once a_1 is produced, `#1` in s_2 is rewritten as a_1 before retrieval and answering. The KG-triple notation is compact, consistent across datasets, and aligned with Wikidata-style supervision in 2WikiMultihopQA and PopQA.

3.3.2 Decomposer Prompt

The decomposer is prompted with a fixed instruction:

> *Decompose the following question into an ordered list of atomic sub-questions. Each sub-question must be answerable in a single retrieval and reasoning step. When a sub-question depends on the answer of an earlier one, refer to that step with a `#N` placeholder. Output one sub-question per line, prefixed with `#<index>:`*

Table 3.1: Representative decomposition patterns produced by the SEEK decomposer.

Dataset	Question type	Decomposition
PopQA	1-hop fact lookup	`#1: George Rankin >> occupation`
HotpotQA	2-hop bridge	`#1: Film X >> director` `#2: #1 >> nationality`
2WikiMultihopQA	comparison	`#1: A >> date of birth` `#2: B >> date of birth`

MuSiQue	compositional chain	`#1: X >> capital`
		`#2: #1 >> mayor`
		`#3: #2 >> party`

3.3.3 Decomposition Examples by Dataset

3.4 Stage 2: Token-Level Confidence Probe

Both the direct path and each sub-question answer use an answer-only prompt:

> *Answer the question with the answer phrase only. No explanation, no preamble, no full sentence — just the answer.*

During greedy decoding, SEEK records the softmax distribution $p_t = \mathrm{softmax}(\mathrm{logits}_t)$ for each generated token t and computes:

$$\sigma^{\text{max-ent}} = \max_t H(p_t), \tag{3.1}$$

$$\sigma^{\text{mean-ent}} = \frac{1}{T}\sum_t H(p_t), \tag{3.2}$$

$$\sigma^{\text{min-lp}} = \min_t \log p_t(y_t), \tag{3.3}$$

where $H(p) = -\sum_v p(v)\log p(v)$ and EOS tokens are excluded. The default diag-

nostic signal is maximum token entropy. In the decomposition path, this entropy controls per-step retrieval. In the v2 verifier, it is used at the query level to decide whether the direct answer is confident enough to rescue a failed decomposition.

3.5 Stage 3: Decomposition-Path Retrieval Gate

Within the decomposition path, retrieval remains sub-question-level. For a resolved sub-question s_i, the answerer first produces a tentative parametric answer $\tilde{a}_i$ and entropy σ_i. The retriever is invoked when:

$$\pi(s_i) = \mathbb{1}[\sigma_i \geq \tau],$$

with $\tau = 0.3$ for max-entropy in the final system. If $\pi(s_i) = 1$, BM25 retrieves top-k passages and the answerer re-answers s_i with context; otherwise $\tilde{a}_i$ is kept. This local gate reduces unnecessary retrieval inside the decomposition path, but it is no longer the final routing mechanism for the whole question.

3.5.1 Original-Question Evidence Union

A decomposed sub-question is often shorter and more entity-centric than the original question, which improves retrieval precision but can remove global clues such as comparison targets or disambiguating conditions. To preserve this context, the final system augments each retrieved sub-question context with evidence retrieved from the original question:

$$C_i = r(s_i, \mathcal{P}, k) \cup r(q, \mathcal{P}, k).$$

The union is deduplicated before answering. This operation does not change the decomposer or train a new retriever; it simply keeps the original-question evidence available when answering each decomposed step. Chapter 4 shows that this is the main improvement from the path-aware baseline to the final system, especially on HotpotQA and

2WikiMultihopQA.

3.5.2 Retriever

The final experiments use an honest open-domain Wikipedia corpus: the shared Elasticsearch `wiki` index over `psgs_w100` with approximately 21M passages, matching DeepRAG's retrieval setting. We use BM25 top-$k = 8$ for HotpotQA, 2WikiMultihopQA, and MuSiQue, and top-$k = 3$ for PopQA. Earlier per-question distractor-pool results are retained only as development and ablation evidence. The final implementation uses a path-aware title policy: title matching is disabled for long original-question direct retrieval and enabled for short decomposed sub-question retrieval.

3.6 Stage 4: Synthesis

The decomposed answer a^{dec} is produced by a synthesis pass over the original question and all (s_i, a_i) pairs. The prompt encodes three answer-shaping rules derived from error analysis:

1. **Comparison questions** return the matching entity, not the compared value.
2. **yes-or-no questions** return "Yes" or "No" based on agreement between sub-answers.
3. **Terse-span rule** strips unnecessary modifiers and keeps answer spans short for EM compatibility.

The implementation also includes a value-to-entity guard for comparison failures: if synthesis emits a value such as a year but the question asks which entity satisfies a comparison, the system maps the value back to the owning entity when this mapping is unambiguous.

3.7 Stage 5: Training-Free Verifier v2

The verifier chooses between the direct candidate and the decomposed candidate. It is intentionally conservative: decomposition remains the default because it improves

multi-hop retrieval, and the verifier switches to the direct answer only when the direct path is confident and the decomposition path shows signs of instability. The final implementation additionally applies a conservative type-aware fallback that corrects a small number of answer-type mismatches without rerunning generation or training a router.

Signals. The verifier uses four query-level signals:

- **Direct-answer entropy** σ^{dir}: low entropy suggests the direct answer may be reliable.
- **Decomposition post-entropy** σ^{dec}: high entropy after decomposition and synthesis suggests the decomposed answer is unstable.
- **Structural validity**: unresolved placeholders, empty sub-answers, or malformed sub-questions indicate decomposition failure.
- **Candidate agreement**: normalised string agreement between a^{dir} and a^{dec} is treated as mutual support.

Rule. The verifier accepts agreement immediately; otherwise it chooses direct when the direct answer is confident and decomposition is unstable, and chooses decomposition in the remaining cases. The deployed thresholds are `direct_entropy_tau=0.4` and `decomp_post_entropy_tau=0.8`. The type-aware fallback is a deterministic refinement: it only changes the final route when answer type, direct entropy, decomposition post-entropy, and entity-fragment signals consistently indicate that the direct answer should be preferred. No dataset name, gold answer, trained classifier, or evidence-label supervision is used.

Evidence support. We also evaluated evidence-aware variants that score whether the final answer or chain answers are lexically supported by retrieved passages. These variants are not part of the final method: they slightly improve HotpotQA but consistently hurt 2Wiki and MuSiQue, showing that lexical support is not a reliable global routing signal for compositional questions.

3.8 Training Protocol

Only the decomposer requires training. The answerer, synthesizer, and verifier are training-free at the routing stage.

3.8.1 Backbone and LoRA Configuration

- **Backbone:** Llama-3-8B-Instruct.
- **Adapter type:** LoRA [30] on all linear modules.
- **Rank, Alpha, and Dropout:** $r = 16$, $\alpha = 32$, dropout $= 0.05$.
- **Precision:** bf16.
- **Context length:** 1024 tokens.

3.8.2 Training Data

Table 3.2: Composition of the 12K decomposer training corpus.

Source	#samples	Decomposition source
MuSiQue	4,000	Gold sub-questions
2WikiMultihopQA	4,000	Evidence-triple conversion
PopQA	4,000	1-step triples
Total	12,000	

The PopQA component teaches the decomposer that a one-step plan is valid. This improves single-hop robustness, but Chapter 4 shows that it also creates negative transfer on some multi-hop datasets. The verifier is introduced precisely because no single hard decomposition policy is optimal across all query types.

3.8.3 Optimisation

LoRA is trained with LLaMA-Factory [31]: effective batch size 16, learning rate 1×10^{-4} with cosine decay and 10% warmup, 3 epochs, and approximately 38 minutes on one A6000 GPU.

3.9 Inference Modes

Two adapter modes were evaluated during development:

Mode B. LoRA is enabled for decomposition and synthesis but disabled for sub-question answering.

Mode C. LoRA is enabled for decomposition and sub-question answering. This is the final default because it outperformed Mode B in pilot experiments.

For final synthesis, the adapter is disabled (**synthOFF**) because the base instruction model produces shorter and more stable final spans.

3.10 Implementation Summary

Algorithm 1 summarises the final inference loop. Code lives in the `seek` module used by the main methodology code.

The next chapter evaluates this final dual-path system against DeepRAG under the honest wiki-21M retrieval protocol and reports the older distractor-pool experiments separately as development diagnostics.

Algorithm 1 SEEK v2 inference for one question q

Require: question q, corpus $\mathcal{P}$, entropy threshold τ, top-k

$(a^{\text{dir}}, \sigma^{\text{dir}}) \leftarrow \textsc{DirectAnswer}(q, r(q, \mathcal{P}, k))$
$\{s_1, \ldots, s_n\} \leftarrow \mathcal{D}_\theta(q)$
$A \leftarrow \emptyset$
for $i = 1$ to n **do**
    $s_i \leftarrow \textsc{Resolve}(s_i, A)$
    $(\tilde{a}_i, \sigma_i) \leftarrow \mathcal{A}_\theta(s_i, \emptyset)$
    if $\sigma_i \geq \tau$ **then**
        $c_i \leftarrow r(s_i, \mathcal{P}, k) \cup r(q, \mathcal{P}, k)$
        $a_i \leftarrow \mathcal{A}_\theta(s_i, c_i)$
    else
        $a_i \leftarrow \tilde{a}_i$
    end if
    $A[i] \leftarrow a_i$
end for
$(a^{\text{dec}}, \sigma^{\text{dec}}, z^{\text{dec}}) \leftarrow \textsc{Synthesize}(q, \{(s_i, a_i)\}_{i=1}^{n})$
$\hat{a} \leftarrow \textsc{Verifier}(a^{\text{dir}}, a^{\text{dec}}, \sigma^{\text{dir}}, \sigma^{\text{dec}}, z^{\text{dec}})$
return $\hat{a}$

Chapter 4 Experiments

This chapter evaluates the final SEEK system under the same honest wiki-21M retrieval protocol used for the DeepRAG comparison. The experiment narrative follows six questions. First, how does SEEK compare with DeepRAG overall? Second, does decomposition actually improve retrieval? Third, is retrieval still necessary after decomposition? Fourth, what role does the entropy gate play? Fifth, why should original questions and decomposed sub-questions use different retrieval settings? Finally, how much headroom remains in the direct-vs-decompose routing decision?

4.1 Experimental Setup

4.1.1 Datasets

We evaluate on HotpotQA, 2WikiMultihopQA, MuSiQue, and PopQA. HotpotQA, 2WikiMultihopQA, and MuSiQue test multi-hop reasoning, while PopQA tests single-hop factual recall and exposes over-decomposition on simple entity-centric questions. To align with DeepRAG [10], all headline results use the first 1,000 questions of each development set without shuffling.

4.1.2 Retrieval Corpus

The main comparison uses the honest open-domain corpus: the shared Elasticsearch `wiki` index over `psgs_w100`, containing approximately 21M Wikipedia passages. We use BM25 with top-$k = 8$ for HotpotQA, 2WikiMultihopQA, and MuSiQue, and top-$k = 3$ for PopQA. SEEK applies a path-aware title policy: the direct path retrieves with the original question over paragraph text only, while the decomposition path retrieves short sub-question queries over title and paragraph text. This policy is dataset-agnostic because it depends only on query type.

The final system further uses *original-question evidence union*. For each decom-

posed sub-question, the evidence context is the union of passages retrieved by the sub-question and passages retrieved by the original question. This preserves the global context that can be lost when a long question is decomposed into short sub-questions.

4.1.3 Metrics and Baselines

We report SQuAD-style exact match (EM) and token-level F1 after standard normalisation. When a dataset provides aliases, the maximum score over aliases is used. The main external baseline is DeepRAG [10], which learns a sub-question-level retrieval policy through binary tree-search rollouts, SFT, and DPO-style Chain of Calibration. Internal comparisons include the path-aware baseline, the original-question evidence union ablation, and the final SEEK system with type-aware verifier replay.

4.1.4 Implementation Details

All SEEK results use Llama-3-8B-Instruct with the 12K three-source LoRA decomposer trained on MuSiQue, 2WikiMultihopQA, and PopQA. The decomposition path uses Mode C, maximum token entropy threshold $\tau = 0.3$, and synthesis with the LoRA adapter disabled. The final verifier uses the original v2 confidence signals plus a conservative type-aware fallback. This fallback changes only the final choice between the direct and decomposed candidates; it does not rerun generation and does not train an additional model.

4.2 Main Results

Table 4.1 reports the final honest wiki-21M results. The final SEEK system combines path-aware title retrieval, original-question evidence union, entropy-gated sub-question retrieval, and the type-aware training-free verifier.

Finding 1: SEEK improves the four-dataset average over DeepRAG. The final SEEK system reaches 40.52 EM and 47.61 F1, compared with DeepRAG's 38.45 EM and 45.75 F1. The average gain is therefore +2.07 EM and +1.86 F1.

Table 4.1: Main results on the first 1,000 questions of each dev set using the honest wiki-21M pool. Each cell reports EM; F1.

Method	HotpotQA	2Wiki	MuSiQue	PopQA	Average
DeepRAG [10]	40.70; 51.50	48.10; 53.30	22.50; 30.40	42.50; 47.80	38.45; 45.75
Path-aware baseline	31.80; 41.25	46.90; 53.51	24.60; 33.55	51.80; 54.19	38.78; 45.62
+ original-question evidence union	34.30; 44.26	49.90; 56.42	**24.80**; 32.87	**52.00**; **54.83**	40.25; 47.09
SEEK final (Ours)	35.50; 46.11	**50.00**; **56.58**	24.70; **33.06**	51.90; 54.71	**40.52**; **47.61**

Finding 2: SEEK wins on 2Wiki, MuSiQue, and PopQA. On 2Wiki, SEEK improves from DeepRAG's 48.10 EM to 50.00 EM. On MuSiQue, it improves from 22.50 to 24.70 EM. The largest gain is on PopQA, where SEEK reaches 51.90 EM compared with DeepRAG's 42.50.

Finding 3: HotpotQA remains the only dataset below DeepRAG. SEEK improves HotpotQA from the path-aware baseline of 31.80 EM to 35.50 EM, but it still trails DeepRAG's 40.70 EM. This makes HotpotQA the main remaining limitation and motivates the analysis in Sections 4.8–4.9.

4.3 Retrieval Recall

Before evaluating final answers, we first ask whether decomposition actually improves retrieval quality. Table 4.2 reports answer-containing hit@5: a query is counted as successful if at least one of the top five retrieved passages contains the gold answer or one of its aliases. Yes-or-no questions are excluded because the strings "yes" and "no" are not meaningful evidence strings.

Decomposition substantially improves answer-containing retrieval on the multi-hop datasets: +30.64 points on 2Wiki, +20.50 on MuSiQue, and +9.99 on HotpotQA. PopQA is nearly unchanged because it is already single-hop. The union column is strongest on all four datasets, showing that decomposition should not discard the original question's global context.

Table 4.2: Answer-containing hit@5 under the final open-domain wiki retrieval condition. Original-query retrieval uses paragraph text only; decomposed sub-question retrieval uses title and paragraph text.

Dataset	N	Original	Decomposed	Union	Decomp − Original
2Wiki	917	31.73	62.38	**65.32**	+30.64
HotpotQA	941	46.55	56.54	**65.25**	+9.99
MuSiQue	1000	16.90	37.40	**41.60**	+20.50
PopQA	1000	68.10	68.70	**73.30**	+0.60

4.4 Retrieval Is Necessary After Decomposition

Decomposition alone does not produce factual knowledge. Table 4.3 compares three decomposition-path settings: never retrieve, entropy-gated retrieval at $\tau = 0.3$, and always retrieve.

Table 4.3: Ablation of retrieval after decomposition under honest wiki-21M. EM is reported for each answering strategy; retrieval rate is measured inside the decomposition path.

Dataset	Never retrieve	Gate $\tau = 0.3$	Always retrieve	Gain	Retrieval rate
HotpotQA	15.7	25.0	25.1	+9.3	92.7%
2Wiki	29.4	47.2	47.1	+17.8	95.2%
MuSiQue	9.6	24.7	25.0	+15.1	90.8%
PopQA	24.2	51.5	51.4	+27.3	99.3%

Retrieval after decomposition is necessary: the gate improves over never retrieve by +9.3 to +27.3 EM. However, the gate and always-retrieve settings are almost identical in accuracy. This positions entropy gating as an efficiency mechanism rather than the main source of accuracy.

4.5 Threshold Sensitivity

Table 4.4 evaluates the final dual-path system under different entropy thresholds. Higher thresholds reduce retrieval, but the average score declines.

Table 4.4: Final SEEK performance across entropy thresholds. Each score cell reports average EM; F1, and retrieval rate is the average decompose-path retrieval rate.

τ	Average EM; F1	Retrieval rate
0.3	**40.52**; **47.61**	94.16%
0.5	40.45; 47.52	91.32%
0.8	39.98; 47.04	85.97%
1.0	39.50; 46.63	82.35%

The chosen threshold $\tau = 0.3$ is accuracy-first. A threshold of 0.8 saves about eight retrieval-rate points but costs 0.54 average EM. The curve supports a conservative interpretation: under honest wiki-21M, most decomposed steps still require retrieval, and tightening the gate eventually hurts multi-hop datasets such as 2Wiki and MuSiQue.

4.6 Path-Aware Title Retrieval

Original questions and decomposed sub-questions are different query types. A short query such as `Houston Baptist University founded` benefits from title matching because the title often identifies the relevant Wikipedia article. A long original question, however, can contain multiple entities and conditions; title matching can overemphasise a stray entity and retrieve the wrong article. Table 4.5 compares title-field policies under the same verifier setting.

Path-aware title retrieval is the most stable option. It keeps the title benefit for short sub-questions while avoiding the HotpotQA drop caused by enabling title matching on long original questions.

Table 4.5: Title-field matching ablation on honest wiki-21M. Each cell reports EM; F1.

Dataset	Off	Global	Path-aware
PopQA	51.2; 53.5	51.5; 53.9	**51.8**; **54.2**
MuSiQue	22.5; 30.5	24.5; 33.1	**24.6**; **33.5**
2Wiki	45.2; 51.3	45.9; 52.5	**46.9**; **53.5**
HotpotQA	31.9; 41.2	30.5; 39.9	**31.8**; **41.2**

4.7 Original-Question Evidence Union

The preceding retrieval results suggest that decomposition improves precision but may lose global context. Original-question evidence union addresses this by adding the original-question top-k evidence to each decomposed step. Table 4.6 reports the effect.

Table 4.6: Effect of adding original-question evidence to each decomposed sub-question context. Each cell reports EM; F1.

Dataset	Path-aware baseline	+ Union	Change in EM
HotpotQA	31.80; 41.25	**34.30**; **44.26**	+2.50
2Wiki	46.90; 53.51	**49.90**; **56.42**	+3.00
MuSiQue	24.60; **33.55**	**24.80**; 32.87	+0.20
PopQA	51.80; 54.19	**52.00**; **54.83**	+0.20
Average	38.78; 45.62	**40.25**; **47.09**	+1.47

The improvement is largest on HotpotQA and 2Wiki, precisely where decomposition can remove disambiguating information from the original question. This supports a key design principle of SEEK: decomposition should create sharper sub-queries, but it should not erase the original context.

4.8 Direct-vs-Decompose Correctness Distribution

Table 4.7 compares answer correctness across the direct and decomposition paths in the final traces. This analysis asks whether decomposition is always better, or whether the two paths solve different questions.

Table 4.7: Answer-level correctness distribution across direct and decomposition paths in the final traces.

Dataset	Both correct	Direct only	Decomp only	Both wrong	Decomp only − Direct only
HotpotQA	192	111	113	584	+2
2Wiki	141	39	361	459	+322
MuSiQue	71	31	184	714	+153
PopQA	350	28	179	443	+151

The distributions show that decomposition is useful but not universally dominant. On 2Wiki, MuSiQue, and PopQA, the decompose-only column is much larger than the direct-only column, confirming the value of decomposition. HotpotQA is different: direct-only and decompose-only are almost tied. This explains why HotpotQA remains difficult and why the final system must preserve both paths. The large both-wrong counts also show that many errors cannot be fixed by routing alone; they require better retrieval, reranking, decomposition quality, or final synthesis.

4.9 Oracle Direct-vs-Decompose Headroom

Table 4.8 reports an oracle that chooses between the direct and decomposed answer using the gold answer. This oracle is not deployable, but it quantifies the remaining headroom in the routing decision.

The oracle gap gives a precise interpretation of the remaining limitation. SEEK has not completely solved direct-vs-decompose routing; rather, it exposes the problem and recovers part of the available headroom with a training-free verifier. HotpotQA has the

Table 4.8: Oracle choice between direct and decompose in the final traces. Current router is the final SEEK verifier output.

Dataset	Direct EM	Decomp EM	Current router	Oracle	Oracle gap
HotpotQA	30.30	30.50	35.50	**41.60**	+6.10
2Wiki	18.00	50.20	50.00	**54.10**	+4.10
MuSiQue	10.20	25.50	24.70	**28.60**	+3.90
PopQA	37.80	52.90	51.90	**55.70**	+3.80

largest gap, +6.10 EM, because neither fixed path dominates.

4.10 Entropy as a Confidence Signal

Entropy remains useful, but it should not be overstated. Across pooled examples, higher answer entropy is associated with a higher error rate, with Spearman correlation around +0.326. This supports max entropy as a sub-question-level retrieval gate. However, low entropy does not guarantee correctness: the model can be confidently wrong. This is why SEEK uses entropy as one confidence signal inside a dual-path system rather than as a single global router.

4.11 Distractor-Pool Development Results

Table 4.9 records the older per-question distractor-pool results. These numbers are useful for diagnosis, but they should not be compared directly with DeepRAG because gold passages are usually guaranteed to appear in the candidate pool.

Table 4.9: Development results under per-question distractor pools. Each cell reports EM; F1.

Dataset	HotpotQA	2Wiki	MuSiQue	PopQA
SEEK development	39.6; 52.4	45.0; 53.7	48.0; 57.1	30.3; 40.2

The gap between distractor-pool and wiki-21M results confirms that retrieval corpus

difficulty is a major confound. The final thesis therefore uses wiki-21M as the headline protocol.

4.12 Summary

The final experiments support four conclusions:

- **SEEK outperforms DeepRAG on average under honest wiki-21M.** It wins on 2Wiki, MuSiQue, and PopQA, while HotpotQA remains the only dataset below DeepRAG.
- **Decomposition improves retrieval, but retrieval remains necessary.** Answer-containing recall improves strongly on multi-hop datasets, and never-retrieve ablations lose 9.3 to 27.3 EM.
- **Entropy is an efficiency signal, not a correctness guarantee.** The gate nearly matches always-retrieve accuracy while saving some retrieval, but low entropy can still be wrong.
- **Direct and decomposition paths are complementary.** Oracle gaps of 3.8 to 6.1 EM show that direct-vs-decompose routing remains an important source of future improvement.

Chapter 5 Discussion

This chapter interprets the final results from Chapter 4. The central lesson is that adaptive RAG should not be reduced to a single retrieve-or-skip decision. In multi-hop QA, the system must also decide when to trust decomposition, when to preserve the original question, and when the remaining error is outside the reach of routing.

5.1 Cost–Accuracy Trade-off versus DeepRAG

DeepRAG [10] and SEEK both operate at a finer granularity than standard retrieve-then-read RAG, but they pay very different training costs.

Table 5.1: Approximate training and routing-cost comparison between SEEK and DeepRAG.

Component	SEEK	DeepRAG
Decomposer data	12K supervised decomposition examples	5K seeds expanded by tree search
Training stages	one LoRA SFT	SFT plus DPO-style calibration
Routing training	none	preference pairs required
Tree-search rollouts	none	exponential branches per hop
Final route	deterministic verifier	learned implicit policy
Interpretability	entropy, type, and stability rules	opaque learned policy

The final results make the trade-off concrete. SEEK reaches 40.52 average EM and 47.61 average F1, exceeding DeepRAG's 38.45 EM and 45.75 F1. It wins on 2Wiki, MuSiQue, and PopQA, while HotpotQA remains below DeepRAG by 5.20 EM. The contribution is therefore not merely a cheaper implementation of DeepRAG; it is a different design point: sub-question-level adaptive retrieval with an explicit direct-vs-decompose verifier rather than an internally learned retrieval policy.

5.2 Why Direct and Decomposition Paths Must Coexist

The answer-level 2x2 analysis shows that decomposition is valuable but not always dominant. On 2Wiki, MuSiQue, and PopQA, the decompose-only column is much larger than the direct-only column, which confirms that decomposition solves many examples the direct path misses. HotpotQA is the exception: direct-only and decompose-only are nearly tied. This means HotpotQA is not simply an always-decompose dataset; the original question often carries comparison targets or disambiguating clues that are weakened by decomposition.

This is why SEEK keeps both paths. The direct path is not only a baseline; it is a context-preserving candidate. The decomposition path is not only a reasoning trace; it is a retrieval strategy that can expose intermediate entities. The verifier is needed because each path fails on different examples.

5.3 Original-Question Evidence Union

Original-question evidence union is the most important methodological update in the final system. Decomposition improves answer-containing retrieval, but short sub-questions can lose the global context needed to interpret the relation correctly. By adding original-question evidence to each decomposed step, SEEK recovers part of this lost context without changing the decomposer or training a new retriever.

The effect is largest on HotpotQA and 2Wiki: HotpotQA improves from 31.80 to 34.30 EM before the type-aware fallback, and 2Wiki improves from 46.90 to 49.90 EM. This supports the thesis motivation directly: decomposition should make retrieval sharper, but it should not force the system to forget the original question.

5.4 Entropy: Useful Signal, Not Correctness Guarantee

Entropy remains useful for deciding whether a sub-question should retrieve. The never-retrieve, gate, and always-retrieve ablation shows that the entropy gate closely

matches always-retrieve accuracy while saving a modest amount of retrieval. However, entropy is not a correctness guarantee. A model can be confidently wrong, especially when it produces a familiar but incorrect entity or answers the first hop of a multi-hop question without recognising the missing relation.

This explains why the final system uses entropy inside a broader verifier rather than as a single hard router. Entropy helps locate uncertainty, but the direct-vs-decompose choice also needs answer type, decomposition stability, and candidate agreement.

5.5 Remaining Headroom

The oracle analysis gives a measured upper bound for better routing. In the final traces, oracle direct-vs-decompose routing reaches 41.60 EM on HotpotQA, 54.10 on 2Wiki, 28.60 on MuSiQue, and 55.70 on PopQA. The current verifier still leaves 3.8 to 6.1 EM of headroom.

This headroom should be interpreted carefully. Oracle routing is not deployable because it uses the gold answer after the fact. It also cannot fix examples where both paths are wrong. The large both-wrong counts, especially on HotpotQA and MuSiQue, show that future gains require not only a stronger verifier but also better retrieval, reranking, decomposition quality, and final synthesis.

5.6 Threats to Validity

5.6.1 DeepRAG Numbers Are Paper Numbers

DeepRAG results are taken from the paper rather than reproduced in the exact same harness. Normalisation and implementation details may differ slightly. To reduce protocol mismatch, SEEK uses the honest wiki-21M pool and the same first-1,000-question evaluation style.

5.6.2 Dataset Slicing

All headline results use the first 1,000 dev questions to align with DeepRAG. MuSiQue dev is sorted by hop depth, so this slice understates full-dev difficulty. Full-dev MuSiQue remains harder, especially for 4-hop questions.

5.6.3 Single Seed and Greedy Decoding

All numbers use greedy decoding without seed averaging. Small differences of 1 to 2 EM should be interpreted cautiously; larger effects such as PopQA's gain over DeepRAG, the 2Wiki improvement, and the original-question evidence union gains are more robust.

5.6.4 Routing Cannot Fix Both-Wrong Cases

The verifier can only choose between two existing candidates. If both direct and decomposed answers are wrong, routing cannot recover the example. This is the main reason oracle routing is an upper bound on routing quality rather than an upper bound on the whole QA system.

Chapter 6 Conclusions and Future Work

6.1 Summary of Contributions

This thesis introduced **SEEK**, a lightweight adaptive RAG framework for open-domain question answering. The main contribution is to frame adaptive RAG as a *decomposition-routing problem*: the system must decide not only when to retrieve, but also when to trust a decomposed reasoning path and when to preserve the original question.

The principal contributions are:

1. **Dual-path adaptive RAG.** SEEK generates both a direct answer and a decomposed answer, allowing the final system to compare original-question reasoning with step-by-step decomposition.

2. **KG-triple decomposition for sub-question retrieval.** The decomposer emits compact `entity >> relation` fragments with `#N` dependencies, making multi-hop retrieval queries explicit and executable.

3. **Entropy-gated sub-question retrieval.** Maximum token entropy provides a transparent and training-free signal for deciding whether each decomposed step should retrieve external evidence.

4. **Original-question evidence preservation.** The final system unions original-question evidence into each decomposed step, reducing context loss caused by overly short sub-questions.

5. **Training-free verification.** SEEK selects between direct and decomposed candidates using confidence, answer type, decomposition stability, and candidate agreement, without classifier training, tree search, DPO, or preference data.

6. **Honest wiki-21M evaluation.** Under the DeepRAG-aligned open-domain pool, SEEK reaches 40.52 average EM and 47.61 average F1, outperforming DeepRAG

on average and winning on 2Wiki, MuSiQue, and PopQA.

6.2 Limitations

The principal limitations are:

- **HotpotQA remains below DeepRAG.** SEEK improves HotpotQA to 35.50 EM, but still trails DeepRAG's 40.70 EM. HotpotQA is the clearest case where direct and decomposed paths are complementary but the current verifier does not fully exploit the headroom.
- **Verifier headroom remains.** Oracle direct-vs-decompose routing still offers 3.8 to 6.1 EM over the current verifier, so routing is not solved.
- **Entropy is not a correctness guarantee.** Entropy is useful for sub-question retrieval gating, but low entropy can correspond to confident wrong answers.
- **Both-wrong cases require better retrieval and synthesis.** Routing cannot fix examples where both direct and decomposed candidates are wrong.
- **Single-seed greedy decoding.** Variance estimates are not reported and would strengthen small-difference claims.

6.3 Future Work

6.3.1 Stronger Verifier Signals

The largest immediate opportunity is improving the verifier. Candidate signals include self-consistency disagreement, top-1 versus top-2 logit margin, hidden-state variance, answer-type compatibility, and lightweight learned calibration from a small validation set. The goal is to approach the oracle direct-vs-decompose headroom without losing SEEK's interpretability.

6.3.2 Better Retrieval and Reranking

The large both-wrong counts show that routing alone is insufficient. Dense retrieval, hybrid BM25+dense retrieval, or reranking over the unioned evidence could improve cases where neither path retrieves enough supporting information.

6.3.3 Length-Conditioned Decomposer

A target-depth control token such as `[K=1]`, `[K=2]`, or `[K=4]` could reduce negative transfer between single-hop and multi-hop data. The verifier or a small complexity predictor could choose the target depth at inference time.

6.3.4 Multi-Sample Decomposition

Sampling multiple decompositions and selecting by final-answer confidence or agreement could reduce wrong-first-hop propagation. This would trade inference cost for robustness without requiring DPO.

6.3.5 Span Calibration

A small post-training or post-processing module for terse answer spans could close EM-only failures while preserving F1. This is especially relevant for verbose entity descriptions and comparison questions.

6.4 Closing Remarks

The final lesson of this thesis is that adaptive RAG should ask two questions: when should the system retrieve, and when should it trust decomposition? SEEK shows that a lightweight system can answer both questions with interpretable inference-time signals. Decomposition improves retrieval, entropy helps control sub-question evidence use, original-question evidence preserves global context, and a training-free verifier makes the final route explicit. The remaining oracle gap leaves a clear path for future work, but the central result is already useful: adaptive RAG can be competitive with heavier learned-

policy systems without hiding its routing decisions inside another trained model.

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Appendices

Appendix A Supplementary Material

This appendix contains supplementary material such as full prompt templates, additional experimental results, dataset statistics, hyperparameter settings, and source code snippets.